Upendra Sai Chaturvedula

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Professional Summary

Entry-level AI & Full-Stack Developer with an MCA and B.Sc. in Computer Science. Hands-on with full-stack and AI-powered applications, plus a few blockchain projects, using Java, Python, React, Node.js, Flask, and LLM APIs (Groq). Built and shipped 8+ end-to-end projects independently, backed by certifications in SQL, Data Analysis, Prompt Engineering, and Anthropic's Claude/MCP tools. Looking for Software Developer, Full-Stack Developer, or AI Engineer roles, available immediately and open to relocating anywhere in India.

Education

MCA | Avanthi Institute of Engineering & Technology (Affiliated with JNTUH)
2024 – 2026 | Status: Completed
Hyderabad

B.Sc. Computer Science | Vivekavardhani Degree College, Kothagudem (Affiliated with Kakatiya University)
Completed 2024 | Warangal | CGPA: 6.51/10

Intermediate & 10th | Sri Nalanda Jr. College (2019) & Singareni Collieries High School (2017)
CGPA: 6.32/10 & 6.7/10 | Kothagudem

Experience

Academic & Freelance Project Experience
Independent Software & AI Developer **2024 – Present**

- 8+ applications built and deployed from scratch, spanning full-stack development, AI integration, and blockchain work, with database design, REST APIs, and security built into each one
- Used Claude and Groq in the dev workflow to prototype faster, debug quicker, and ship sooner

Technical Skills

Languages: Java (OOP, Java 8+), Python, JavaScript, SQL, PL/SQL, C++

AI & LLM: Groq API, LLM Integration (LLaMA), Prompt Engineering, Cloud API Integration

Web Development: React.js (Vite), Node.js, Express.js, Spring Boot, Spring Security, Flask, JSP, Servlets, HTML5, CSS3, REST API Design

Database: MySQL, PostgreSQL, CRUD Operations, Database Design

Architecture & Tools: Software Architecture, Fault-Tolerant System Design, Apache Tomcat, Git, GitHub, Docker, Postman

Security & Concepts: Counting Bloom Filters (CBF), Cryptographic Verification, Data Security & Encryption, Public Verifiability

Certifications

- SQL and Relational Databases 101
- Data Analysis Using Python
- Prompt Engineering for Everyone
- **Deloitte:** Cyber Security Simulation, Data Analytics Simulation
- **Anthropic:** Claude 101, Claude Code 101, Claude Code in Action, Claude with the Anthropic API, Introduction to MCP, MCP Advanced Topics
- **Udemy:** Complete Prompt Engineering for AI Bootcamp (2026), Complete Full-Stack Web Development Bootcamp

Key Strengths

- **AI-Assisted Efficiency:** Claude, Groq, and prompt engineering keep prototyping and development moving fast
- 8+ projects, built independently and deployed one by one, alongside MCA coursework
- Most of the real problem-solving came from debugging deployment issues and getting APIs to actually talk to each other
- Quick learner. Self-taught new technologies (Groq API, Motoko/ICP, RAG) through independent research
- Cares about writing code that stays clean and scales, and that someone else could maintain without a headache

Projects

AI Projects

Panda AI - Multi-User AI Assistant Platform
Flask, Python, Groq API, REST APIs
A text chatbot wasn't enough. I wanted something with real web search and file understanding, so I built this instead of relying on a single paid API.

- Multi-user AI assistant with real-time search grounding and multimodal file processing: PDF, image, audio, video
- Fallback architecture across 3 Groq API keys plus OpenRouter, so it stays up and costs nothing to run
- Kept the REST API stateless, so multiple users can work at once without any drop in performance

Universal AI-Powered Resource & Budget ERP System
Java, Groq LLM, REST API
AI can do more than run a chatbot. This explores how it can support real business decisions, like knowing when to reorder stock or flagging a suspicious transaction.

- Multi-domain ERP system with an AI engine handling business insights, auto-reordering, and fraud detection
- Built the REST API and browser dashboard to support concurrent multi-user sessions across 4+ business modules

Java Projects

Secure Cloud Migration & Verifiable Deletion System
Java 8, Counting Bloom Filters, Cryptographic Verification, MySQL, Apache Tomcat
This was my MCA major project at Avanthi Institute, built to solve a real cloud security problem: migrating data safely between providers while guaranteeing permanent, verifiable deletion, without relying on a third party.

- Cloud-to-cloud data transfer and verifiable deletion framework, built on Counting Bloom Filters (CBF), which removes the need for a Trusted Third Party (TTP)
- Added cryptographic confidentiality, integrity validation, and public verifiability protocols so malicious server activity can actually be caught

Employee Leave Management System
Java 17, Spring Boot 3, Spring Security, JWT, Spring Data JPA, MySQL/H2
Wanted real hands-on practice with the framework companies keep asking for, so I built a full backend instead of just watching tutorials.

- REST API where employees apply for leave and only their own manager can approve or reject, secured with JWT and role-based access
- Optimistic locking and balance re-checks keep approvals safe, verified with 28 JUnit 5 and Mockito unit tests

Web Development Projects

WaveCoin - Decentralized Token Application
Motoko, Internet Computer (ICP), DFX
Blockchain always felt like a black box, so I built this to see how minting and transfers actually work under the hood, instead of just reading about crypto.

- Custom cryptocurrency logic for minting and transferring tokens on-chain
- Deployed the canisters and built a frontend for real-time balance tracking

Advanced React Blog Platform
React.js, Node.js, Express, PostgreSQL/MongoDB
Basic CRUD wasn't enough of a challenge, so this became a deeper dive into real authentication and session handling in a full-stack app.

- Full CRUD app, with secure authentication and session handling built in

Additional Projects

Also built an AI Blog Platform with AI content generation and a RAG chatbot, using React and the Groq LLaMA API. There's an Inventory Management System in Java Swing for real-time stock tracking, and an E-Commerce Console App in Java and MySQL that simulates a full shopping and wallet system. And PeakNFT, an NFT minting platform built with React.js and Solidity/Motoko, using decentralized storage for metadata. Full code for all of these is on GitHub at github.com/Upendrasai1101.